Inclined Plane Force Problems

1. A 25 kg luggage trolley rolls down a straight airport access ramp inclined at 20° to the horizontal. Rolling resistance is a constant 20 N up the ramp. Take $g = 9.81\,\mathrm{m\,s^{-2}}$ throughout.

   (a) Calculate the normal reaction from the ramp on the trolley.

   (b) Calculate the trolley's acceleration down the ramp.

2. A 40 kg equipment trolley is being hauled up a loading ramp inclined at 15° to the horizontal. A cable pulls it with a force of 180 N parallel to the ramp. Rolling resistance is 25 N down the ramp.

   (a) Calculate the normal reaction from the ramp on the trolley.

   (b) Calculate the trolley's acceleration up the ramp.

3. A 30 kg equipment sled is being lowered down a hillside by a rope held by a rescue team uphill. The rope's tension is 50 N, acting parallel to the slope. The normal reaction from the ground is 255 N, and the sled accelerates down the slope at $0.80\,\mathrm{m\,s^{-2}}$. Friction acts up the slope.

   (a) Calculate the angle of the hillside to the horizontal.

   (b) Calculate the magnitude of the frictional force.

4. A 50 kg delivery crate moves up a loading ramp inclined at 10° to the horizontal. A winch pulls parallel to the ramp, while a hold-down strap presses the crate perpendicular to the ramp. The normal reaction on the crate is 510 N, and resistance to its motion is 35 N down the ramp. It accelerates up the ramp at $0.80\,\mathrm{m\,s^{-2}}$.

   (a) Calculate the force exerted by the hold-down strap on the crate.

   (b) Calculate the pulling force exerted by the winch.

Answers

1. **(a)** 230 N **(b)** 2.56 m s^{-2} down the ramp

2. **(a)** 379 N **(b)** 1.34 m s^{-2} up the ramp

3. **(a)** 29.9° **(b)** 72.9 N up the slope

4. **(a)** 27.0 N into the ramp **(b)** 160 N up the ramp